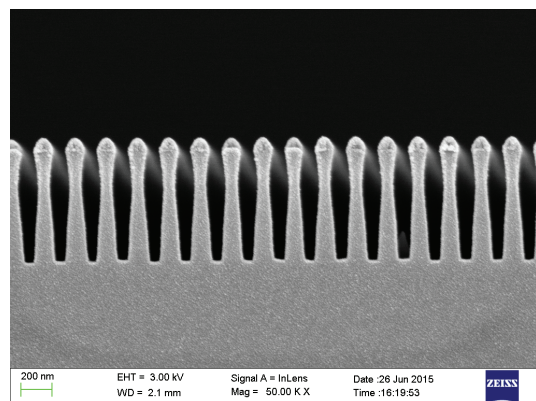
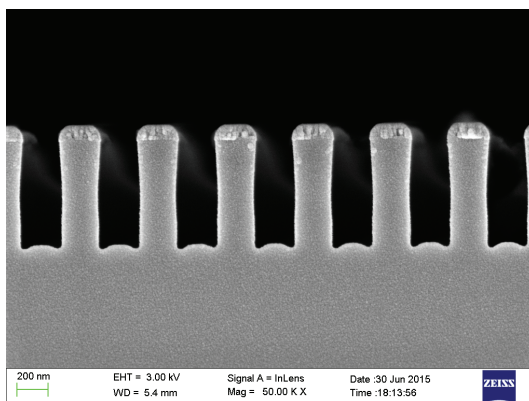


Etching Process Development of SiO_2 Etching Using Inductively Coupled Plasma

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Abstract

The etching characteristics of SiO₂ have been studied systematically using inductively coupled plasma. The etch rate, selectivity, sidewall angle, polymer deposition, micro-trenching and surface morphology have been investigated against the different process parameters for two binary gratings having periods of 200 nm and 500 nm with a fill factor of 0.5 for the both of them. Chromium was used as a hard mask and deposited on top of a fused silica substrate by physical vapor deposition technique. Electron beam lithography was used to pattern the sample and dry etching technique to transfer the pattern into the chromium layer.

The inductively coupled plasma etching of SiO₂ was carried out using SF₆, CF₄ and CHF₃ reactive gases with the different combination of O₂ and Ar additives. Other investigated process parameters were RF power, ICP power, chamber pressure, gas flow rate and substrates backside cooling. The etch rate was expected to be below 100 nm min⁻¹ in order to etch precisely within the sub-micrometer scale. The process parameters were optimized systematically by characterizing the etched outcomes through a scanning electron microscope. An excellent result was obtained using the CHF₃ reactive gas with the 60% Ar additive. Flat and vertical sidewall was achieved with the etch rate of 71.9 nm min⁻¹. The optimized parameters were: the CHF₃/Ar gas flow rate of 10/15 sccm, the RF power of 20 W, the ICP power of 600 W, the chamber pressure of 3 mTorr, the He pressure of 5 Torr (backside cooling). However, this process was optimized for the submicron structures and beyond that scale different optimized solutions are presented.

Preface

First of all, I would like to express my deepest gratitude and honor to my supervisors Prof. Markku Kuittinen and Ph.D. Janne Laukkanen for guiding me throughout my Master's thesis and providing me the best possible facilities. They taught me a lot of theoretical and practical things related to the micro- and nanofabrication, patiently answered thousands of stupid questions and allowed me to work freely in the cleanroom environment.

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In the field of micro- and nanofabrication, etching is considered as the removal of material in a selective manner. Until the late 1970s, isotropic chemical wet etching was the most preferred pattern transfer technique [1]. The first demonstration of oxygen plasma ashing of a polymer-based photoresist film was introduced by Irving in 1968 [2]. Later in 1971, he revealed and patent the experiment of plasma etching of silicon using CF_4 gas [3]. Then in the early 1970s, both plasma ashing and plasma isotropic etching technology were successfully used in semiconductor manufacturing companies [4]. This revolutionary extension of the etching technique brought drastic changes in the semiconductor manufacturing process. Since then, plasma etching technologies have been developing abruptly. Hosokawa *et al.* in 1974 first investigated the etching characteristics of fluorine and chlorine-containing gases instead of Ar for various materials using a parallel plate reactor [5]. His first motivation was to increase the etch rate of different materials [4]. He also developed an optimized process to etch SiO_2 using C_3F_8 gas in 1975 [6]. Now his distinct technique is known as reactive ion etching (RIE). RIE is much broad range of plasma based dry etching technique [7] and yet, it is the most widely used pattern transfer technique [8].

RIE is perfectly capable to high anisotropic etch of a sub-100 nm enhanced structure [8]. However, until the mid-1990s, one had to struggle three major problems related to RIE: low etch rate, low selectivity and inability to maintain high-aspect ratio etch [9, 10]. Deep reactive ion etching (DRIE) was critical with RIE for these problems. High density plasma sources provide the solution of these problems. Helicon plasma, electron cyclotron resonance (ECR) and inductively coupled plasma (ICP) are three common high density plasma sources [11]. Recently devel-

oped ICP–RIE is one of the most popular etching techniques which cleverly solved the major limitations of RIE. Nowadays it widely used for DRIE and caught the attention of many researchers [12]. However, there is plenty of scope in this etching technique that yet has to be developed.

On the other hand, SiO_2 is an important material, especially in the field of photonics because of its outstanding properties, such as transparent in a wide wavelength range, strong damage resistance to laser irradiation, high thermal and chemical stability [13]. However, these properties make it difficult to fabricate. The major challenges are low etch rate and selectivity [14]. With RIE, the etch rate is only about 50 nm min^{-1} , where, ICP can deliver 10 times higher etch rate with the better selectivity and anisotropic etch [8]. Numerous studies have been reported on SiO_2 etching using ICP [15–21] but no comprehensive optimized process exists. This is obvious because, the stability and reproducibility of the ICP etching are strongly dependent on the reactor configuration, geometry, gas dynamics, plasma discharge method, microwave frequencies and materials [4]. Therefore, each ICP system has to be calibrated and optimized separately. Our main motivations were to realize the etching characteristics and develop a typical optimize process so that one can fabricate better components made of SiO_2 with the aid of novel ICP.

In this study, the whole experiment was done in cleanroom environment. The ICP etching characteristics of SiO_2 were studied by investigating etched profiles of two binary gratings against different process parameters. Chromium was used as a mask material and deposited on top of a fused silica substrate by physical vapor deposition technique. Electron beam lithography was used to create the pattern on a resist and dry etching technique to transfer the pattern into the chromium layer. The sample was then etched by ICP with different process parameters. Etched profiles were characterized with a scanning electron microscope (SEM). Therefore, Chapter 2 deals with the basic principles of these fabrication techniques. The theoretical aspects of the different process parameters which have been investigated during ICP etching are also discussed. Chapter 3 introduces the fabrication method and experimental procedures. The aim of this chapter is to provide all the possible information so that one can repeat the fabrication method and the experiment. Chapter 4 includes results and discussions. The detailed investigation method, effects of different process parameters with comparisons, optimization method, limitations and some important SEM images are presented. Finally, Chapter 5 draws the conclusions.

There are numerous ways of micro– and nanostructure fabrication. In this study, we have utilized the electron beam lithography technique to create the desired pattern onto a electron sensitive polymer. Dry etching technique has been employed to transfer the pattern into the chromium layer and then into the fused silica substrate. In this chapter, we will briefly discuss the basic principles of these techniques and the effects of different process parameters in inductively coupled plasma etching.

2.1 Patterning

Lithography is a technique typically used to create desired pattern on a suitable polymer which is often called resist. There are different type of lithography techniques involved in micro– and nanofabrication, such as photolithography, electron beam lithography (EBL), nanoimprint lithography (NIL), focused ion beam lithography (IBL), interference lithography, X-ray lithography, gray–scale lithography and near–field holography [22, 23]. Each of them has various strengths and weaknesses. The choice of material, profiles structure, optical functionality of the component, flexibility and expenses can limit the choice of method that can be used. The most common fabrication process involves photolithographic technique. However, EBL widely used nowadays in the area of nanotechnology because of its resolution at the molecular scale. The resolution and accuracy of the EBL system is far better than the photolithographic system. Less than 5 nm order feature size can be achievable by the EBL system [24–26]. In this study, an EBL system was used to pattern the desired gratings on a resist material.

2.1.1 Electron Beam Lithography

A basic EBL system consists of an electron gun, electron optical column, alignment system, blanker, electron detector and a vacuum chamber containing a laser controlled stage for accurately positioning of the substrate [27, 28]. A schematic diagram of a typical EBL system is shown in Fig. 2.1. A vacuum system provides proper environment inside the column. The electrons are emitted from the electron gun and propagate through several additional stages in the electron optical column that performs specific beam modification processes to get required current, spot size and focus. The emitted electrons are accelerated by a voltage and then alignment system centers the beam inside the column. The magnetic lenses are used to focus the beam, the blanker to turn the beam on and off depending on requirement and the deflector to scan the beam. Finally, an objective lens focuses the beam onto the substrate [8, 29]. As the electron beam penetrates into the resist, forward and back scattering occurs. Forward scattering occurs due to the electron-electron interaction and it causes the broadening of the line width. Forward scattering can be controlled by a very high accelerating voltage if necessary [29]. Backscattering occurs due to the interaction of electrons and atoms of the target substrate. The scattered electrons propagate onto the unintended portion of the resist and it broadens the groove width inconsistently. This phenomenon is known as proximity effect [28].

Typically, high resolution and accuracy can be obtained with the small size of focused electron beam [29], which can be achieved from a small size of electron source [8]. However, there are other factors that can affect the final resolution of the fabricated pattern, among them beam current, exposure dose, properties of resist, electron scattering (proximity effect) are the most substantial. Spot size can be maintained by the operating voltage, higher operating voltage provides smaller spot size. The groove width is mostly influenced by the forward scattered electrons with the low exposure dose. At higher dose, backscattering dominates and, as a result, increases the proximity effect. Therefore to achieve the smaller groove width, higher beam energy and lower exposure dose are required [31, 32]. The proximity effect is particularly severe for nanostructure fabrication and there are numerous researchers studying this effect and developing different correction methods, such as modifications of the pattern data and compensation of the exposure dose [33, 34].

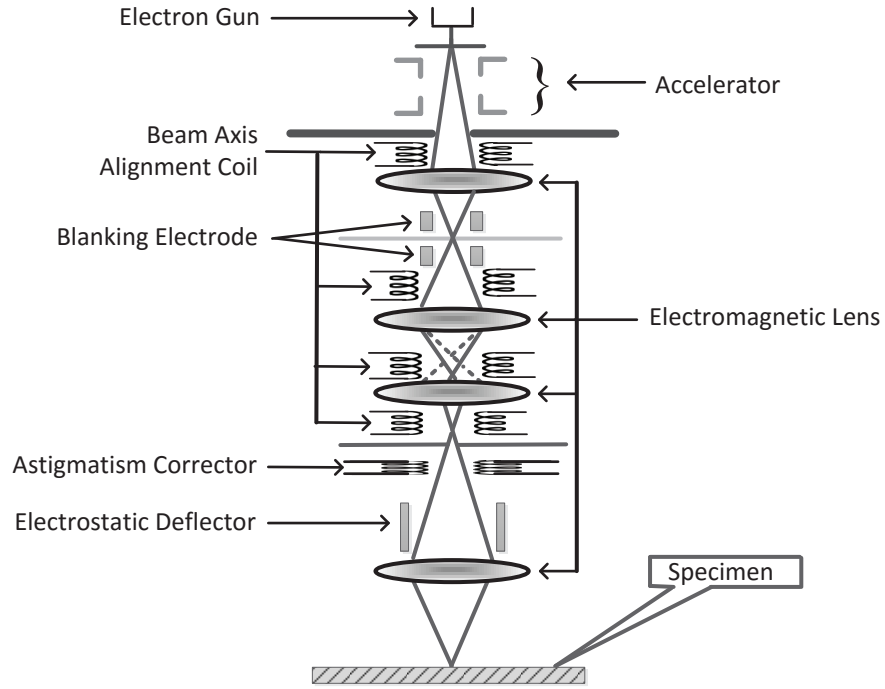


Figure 2.1: Schematic diagram showing major components of an electron beam lithography system [30].

2.1.2 Resist

Resist is a polymer which is chemically sensitive to the energy of incident electron or photon beam. Chemically changed exposed area of the resist is dissolved in a developer solution to obtain desired pattern. According to their response to the energy of the electrons, the resists are divided into two types. The resist which is more soluble to the developer liquid after expose is called positive resist and which is less soluble is called negative resist. The resist should have high resolution, high sensitivity to electrons or photons and excellent etching resistance. The developer characteristics, concentration and development time also play an important role on the resolution of the structure. After patterning and developing the resist, the pattern can be transferred to the substrate by different etching processes.

2.2 Hard Mask

The resist material is not durable enough for deep etching of the substrate. In that case, an extra layer of hard material is deposited between the resist and substrate. The choice of hard material depends on the substrates material, structure and fabrication techniques. However, the common property of the mask is high selectivity. The ratio of the mask to the substrate etch rate is defined as selectivity. Metal, like Chromium, Aluminum, Nickel, Titanium, etc. can be used as a hard mask for dry etching of SiO_2 . In this study, the chromium was used as a hard mask material and it was evaporated on top the substrate by physical vapor deposition (E-beam evaporator) method.

2.2.1 Thin Film Deposition

Thin film deposition is one of the most important steps in micro- and nanostructure fabrication. It can be deposited by Chemical Vapor Deposition (CVD), Physical Vapor Deposition (PVD), Atomic Layer Deposition (ALD), oxidation, spray coating and plating [8,35]. These can be divided into several branches. PVD and CVD are the most common techniques used in fabrication process. However, ALD and PVD are commonly used for the mask material deposition [36]. Different deposition techniques provide different physical structures which is very important factor, because it affects the optical property and further fabrication process of the film [37,38].

2.2.2 Physical Vapor Deposition

In PVD systems the deposited material can be transferred from the source to the wafer by evaporation or sputtering. In evaporation, the source is placed in a crucible, typically located at the bottom of a vacuum chamber and heated up to the evaporation temperature. Various techniques are used to heat the source material. Common ways are high current induction coils wound around the crucible or the electron beam bombardment of the material surface (e-beam evaporator). The technique is mainly used to deposit metals and dielectric materials [39]. A cross-section of a simplified e-beam evaporation chamber is shown in Fig. 2.2.

In the sputtering system, a target of the source material is bombarded with high-energy inert ions, like Argon. As a result, atoms or clusters are removed from the source surface and ejected towards the wafer and consequently deposited on top of

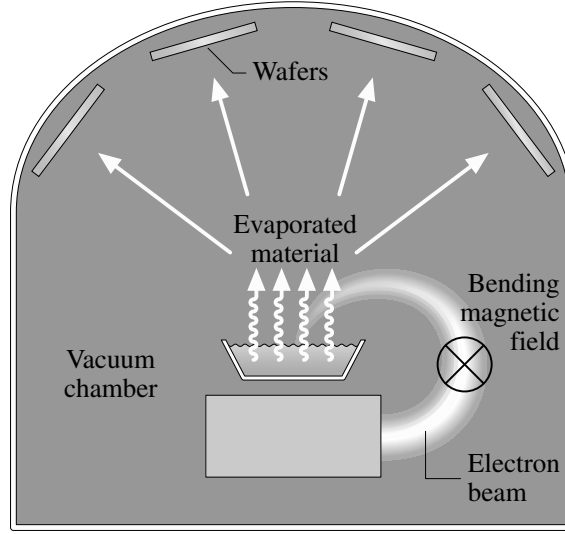


Figure 2.2: Schematic of a typical Physical Vapor Deposition (E-beam heating) system [39].

the wafer surface. The plasma and ions bombardment are produced by direct current (DC) or radio frequency (RF). In a typical system, the target located or attached to bottom of the top electrode and the wafers are placed on top of the bottom electrode. Sputtering can be used for metals, dielectrics, alloys and typically all kinds of compounds [39].

2.3 Etching

Etching is a method in which material is selectively removed in order to get desired pattern. The pattern transfer from resist to the substrate can be done by different etching techniques. Basically there are two types of etching processes, wet etching and dry etching. The wet etching is an isotropic process where liquid chemical etchant is used to etch substrate through chemical reaction. On the other hand, the dry etching is an anisotropic process, where ions and radicals are used to etch the substrate physically or chemically or combination of both [9]. Typically, selectivity and anisotropy are the two parameters that characterize all the etching techniques [8]. Dry etching is widely used in micro- and nanofabrication for its

highly anisotropic characteristic. Ion beam etching (IBE), chemically assisted ion beam etching (CAIBE), ion beam milling (IBM), chemical plasma etching (PE), reactive ion etching (RIE) and reactive ion beam etching (RIBE) are some common dry etching techniques [8, 40]. Among them RIE is the most widely used etching technique in micro- and nanofabrication. More advance RIE system consists of a high density plasma source, like ICP.

2.3.1 Reactive Ion Etching

RIE system uses plasma as a source of reactive species which etches the target material. Plasma is formed by applying a high frequency electric field between two electrodes that results the oscillation of free electrons which are then collide with the gas molecules and causing them to ionize. The region near the cathode establishes a higher field than the anode because it is not grounded. This difference of the electric field accelerates ions toward cathode. As a result, a sample experiences the bombardment of energetic ions that placed on the cathode [8]. Also the applied electric field accelerates the ions bombardment. However, sputtering is not the dominant factor in the RIE etching. The etching is mostly caused by the chemical reaction, because the most RIE process uses chemically reactive and erosive gases [8].

2.3.2 Inductively Coupled Plasma Etching

Low etch rate, low aspect ratio and poor selectivity are the main three limitations of the RIE system. These problems are substantial for deep RIE etching. Solution is to utilize a high density plasma in RIE system, like ICP. A cross-section of such an ICP-RIE (Plasmalab 80+ICP 65) unite is illustrated in Fig. 2.3. In ICP, a coil operates typically at 13.56 Mhz and its power is coupled into the plasma chamber from outside [8, 9]. The sample stage is connected to an auxiliary RF power source to enhance the plasma production. The magnetic field reduces the charge difference between sheath regions significantly which is then reduces the DC bias and hence the ion energy [9]. The RF source in the upper part of the chamber generates the plasma and the auxiliary RF power controls the bias voltage, that is, the ion energy. Thus the ion energy and density are independently controllable in the ICP system. As a result, an ICP system can produce very high density plasma with low ion bombardment energy at low pressure. Hence the ICP resolves the limitation of the RIE and can produce more directional etching with very high etching rate and selectivity.

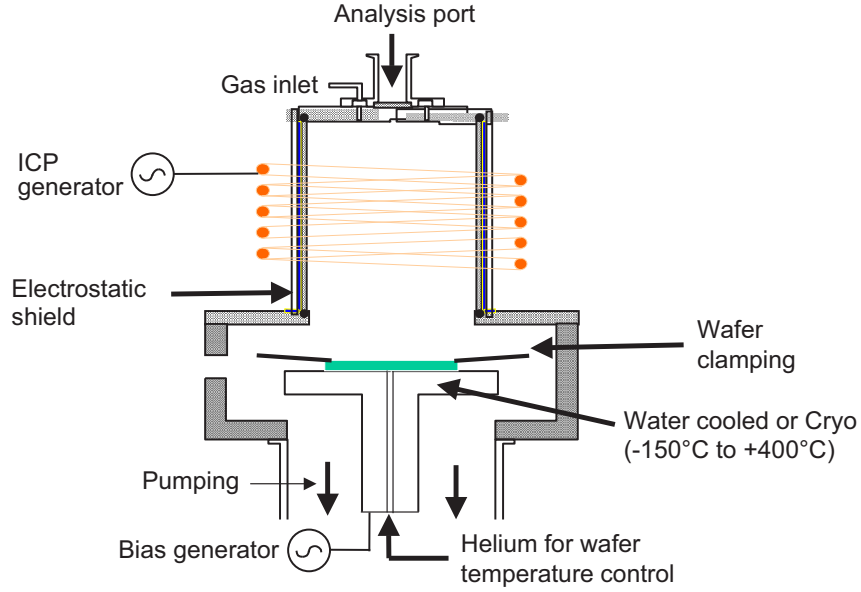


Figure 2.3: Schematic of a ICP-RIE system [41].

The conventional RIE or ICP-RIE resolved many difficulties but still they have some critical issues related to the profile quality, like etch uniformity, control of the structure, damage of the substrate, etc. Etch rate uniformity in both wafer to wafer and within a wafer is critical in most applications. Typically, 3% non-uniformity within a wafer is allowed [42]. The uniformity can be optimized by the reactor design, choice of chemistry and process parameters. Various deformed profiles structure can be found with plasma etching, some of these are illustrated in Fig. 2.4. All of these issues must be considered in evaluating an etching outcome and during the optimization process.

Variation of the etch rate depending on the surface area of the material to be etched is known as a loading effect. This effect observed when the active species in the plasma depleted rapidly by the reaction with etched material [42]. It effected by the lifetime of the active species as well as the chamber volume and feed gas flow rate [42]. The effect is stronger when the etching process is dominated by the chemical etching with neutrals and weaker when the etching is driven by the ions [43]. Three types of loading effect can be found. Macroloading, where the etch rate decreases with increased amount of etched area. Microloading, where the etch rate is lower at the region of higher pattern density. If the etch rate slows down

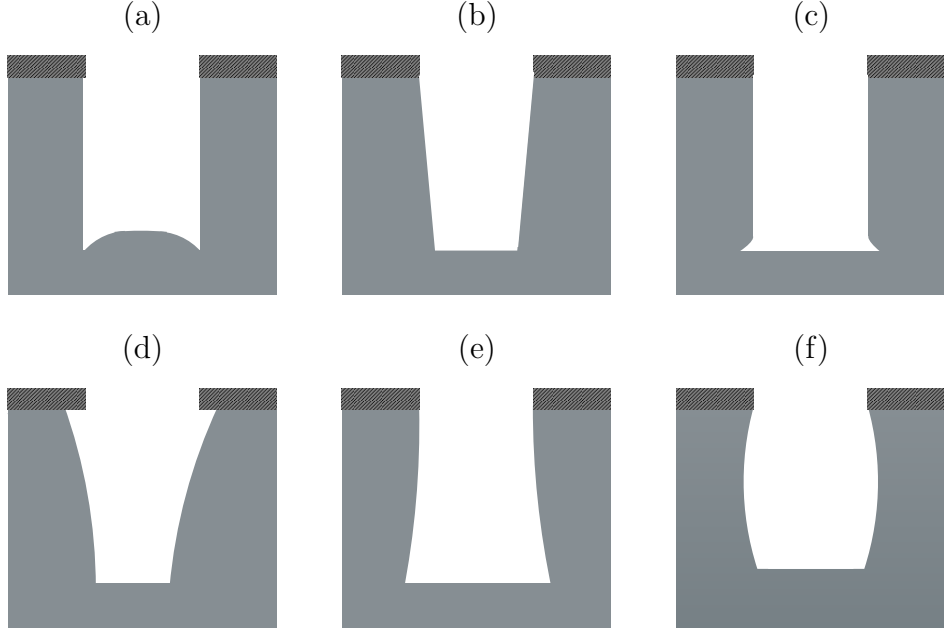


Figure 2.4: Plasma etching profiles: micro-trenching (a), tapered (b), notching (c), undercutting (d), over-cutting (e) and bowing (f).

with the decreased feature size, then the effect is known as aspect ratio dependent loading or RIE lag effect.

When the etch rate is higher near the base of a sidewall than the center of a trench due to enhanced ion flux along the sidewall, then the phenomenon is known as microtrenching effect. Reflection of high energy particles from a slanted mask or sidewall travels towards the base of a sidewall resulting in the higher etch rate [44]. Negative charging of the sidewall also deflects the ions toward the sidewall and causes the microtrenching [45]. The local charging effect and as well as the microtrenching can be reduced by deliberately introducing flux of electrons into the trenches [8, 46]. Slanted sidewall of the mask also plays significant role in the initial development of the microtrenching [44].

Notching or footing is lateral etching at the foot of etched structure typically found in silicon-on-insulator substrates, where a thin insulator (SiO_2) is sandwiched between a thick silicon wafer as a etch stop layer to have the same etch depth for all feature sizes [8]. The notching effect is observed if the substrate is over etched.

2.4 Effects of Process Parameters in ICP etching

Different process parameters in ICP etching affect the profile quality, etch rate and selectivity depending onto the mask and etched material, reactor condition and most importantly the structure of the component. Therefore, process parameters has to be investigated thoroughly in order to optimize the ICP etching. RF power, ICP Power, chamber pressure, flow rate of reactive gases, substrate temperature and gas composition are some typical changeable process parameters. Effects of these parameters are briefly described below.

2.4.1 RF Power

The auxiliary RF power or simply the RF power of a ICP system increases the electron energy and hence, increases ionization probability. As a result, etch rate increases with the higher RF power. Also the increased electron energy provides higher self-bias which accelerates the ions toward the cathode and increases the ions bombardment energy. Consequence is higher sputtering, which is a physical process and increases the etch rate but reduces the selectivity. However, highly directional ions propagation toward the cathode improve the anisotropy of the etch. On the other hand, the main RF power or ICP power controls the plasma density and as we have discussed before, the etching rate and selectivity are improved significantly with higher ICP power. However, excessive ICP power can lead to downward trend in the etching rate because the mean free path of the molecules becomes then shorter [15].

2.4.2 Chamber Pressure

The plasma etching is usually held in low pressure condition and the chamber vacuum is typically $10^{-3} \sim 10^{-1}$ Torr [8]. Chamber pressure affects the mean free path of the ions. Higher pressure leads to more collisions and therefore the directionality of the etching is compromised. Thus, the lower chamber pressure improves the etching rate and anisotropy of the etching. It also helps to remove the etched and volatile products from the chamber. However, the plasma does not start or sustain at too low chamber pressure.

2.4.3 Flow Rate of Reactive Gases

Generally, higher flow rate of reactive gases means more ions in the chamber and provides more etching. However, the number of ions can be decreased if the plasma discharged power is not increased accordingly and this will lead to drop of etching rate [8].

2.4.4 Substrate Temperature

High temperature enhances the chemical process and also helps to remove the volatile products from the etched surface. However, selectivity is decreases in the case of photoresist mask at the higher temperature. It also reduces the anisotropy of the etching, because the chemical process is by nature isotropic. However, highly anisotropic SiO₂ etching can be possible using SF₆ reactive gas at below 0° C temperature [19]. On the other hand, low temperature can lead to re-deposition of etch products on the sidewalls during SiO₂ etching with fluorocarbon plasma [19]. If the substrate needs to be etched for long time, then it requires suitable amount of backside cooling. Due to long time ion bombardment onto the substrate, the temperature increases gradually and that can change the characteristic of etch chemistry.

2.4.5 Gas Composition

Halogen compounds are mainly used to etch inorganic and metallic material. However, additive like O₂, Ar, He, N₂ are often used to affect the etching chemistry. All of these gases affect the etch rate, selectivity, passivation and surface roughness. For instance, addition of O₂ increases the etch rate but generally reduces the selectivity. Depending onto the etch chemistry it also helps to remove deposited polymer, reduces the surface roughness, affects the both sidewall passivation and micro-trenching. Inert gas, like Ar is added to enhance the physical process. Ionized Ar provides electrons to the plasma and helps to stabilize it [47]. It also affects the sidewall passivation. However, too much Ar addition causes excessive physical sputtering which can lead to then reduction of the selectivity and etch rate. The common plasma gases for the SiO₂ etching are CF₄, CHF₃, C₂F₆ and C₃F₆. The dominant reactive species are CF_x which produces the etched products SiF₄, CO and CO₂ [9].

Experimental Procedures of SiO₂ ICP Etching

In this chapter, we will discuss detailed experimental procedure and working principles of the equipment that was used to develop the etching process of SiO₂ using ICP. First, we will discuss the sample preparation and process of the mask fabrication. Then we will discuss the experimental procedure of the SiO₂ etching using ICP. Finally, we will present the characterization technique through the help of a scanning electron microscope. All the used tools and equipment of the experiment are listed in Appendix A.

3.1 Sample Processing

Each investigated samples consists of two binary gratings having periods of 200 nm and 500 nm with a fill factor of 0.5 for the both of them were fabricated and etched down to different grating depths in the range of nanometer to sub-micrometer scale by ICP. Total 144 samples were prepared in this study and among them 32 samples were fabricated with the chromium mask thickness of 100 nm and the rest of them were 50 nm. A set of 16 samples were fabricated at the same time on a wafer made of fused silica having the diameter of 100 mm and thickness of 0.5 mm. Then the wafer was cut into 16 pieces accordingly with the automatic dicing machine. In the following sections, we will briefly discuss the fabrication procedure of the samples. The different steps of the fabrication method are illustrated in Fig. 3.1.

3.1.1 Wafer Cleaning

The wafer made of fused silica required careful cleaning to remove residues or dust particles from its surface. First, it was placed in a beaker with pure acetone and

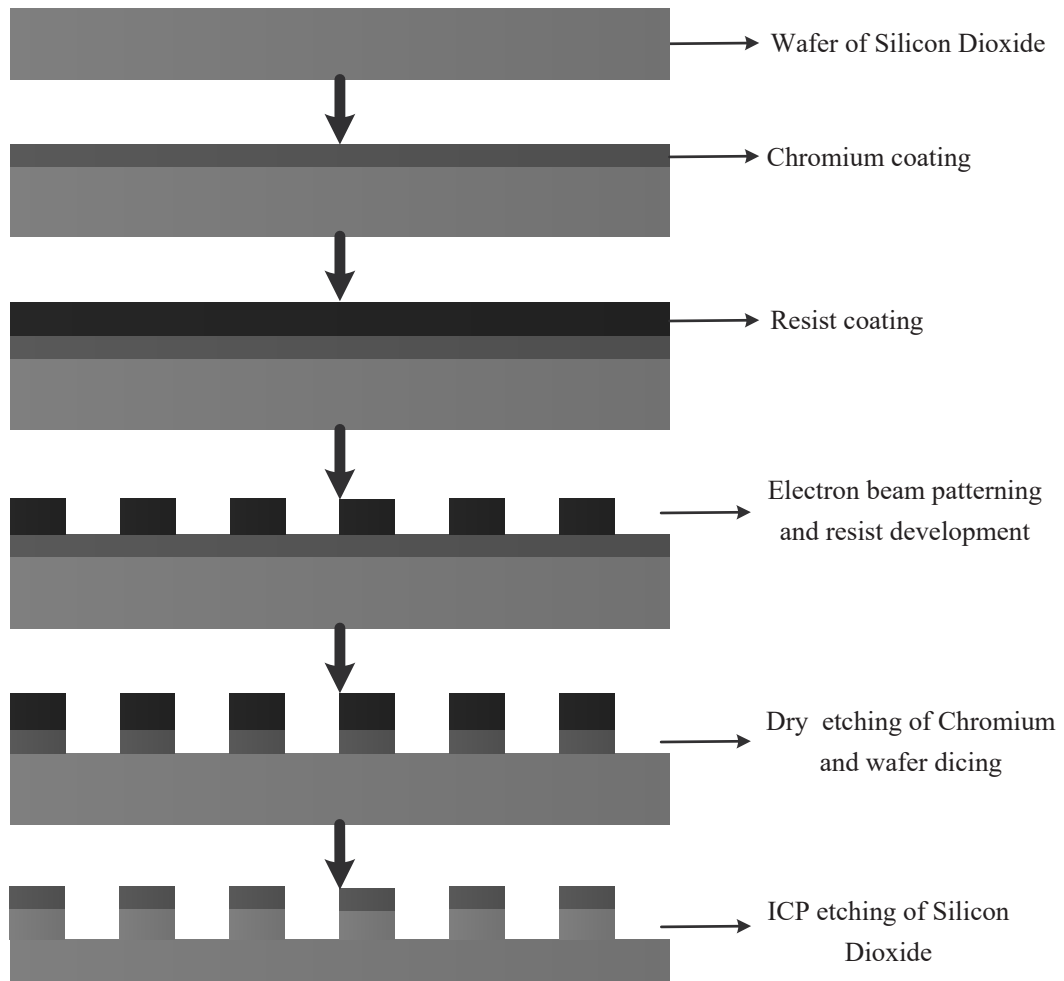


Figure 3.1: Schematic diagram of the fabrication Method.

then moved into an active ultrasonic washing unit for 3 minutes. Then the wafer was placed in a beaker with isopropanol and it was shaken by hand for 2 minutes to remove remain residues. Finally, the wafer was washed and dried with Nitrogen blow.

3.1.2 Chromium Evaporation

Chromium was deposited on top of the wafer by the PVD method. A Kurt J. Lesker LAB18 system was used in this purpose. First, the chamber was opened by selecting ‘Start PC Vent’ in the ‘Vacuum’ option on user control interface of the system. It

required a few minutes to open the chamber. Then from the ‘Deposition’ option we opened the shutter of the source container and checked the amount of chromium. If necessary some 99.98 % pure chromium crystals was added into the container. After that the wafer was carefully placed on the sample stage, then the chamber door was closed. ‘The PC Pump’ was selected from the ‘Vacuum’ option to start the pump down process. It was required around 3:30 hours to reach below the recommended vacuum level of 5×10^{-7} Torr. Since we have used the e-beam deposition, we turned on the high voltage source and then selected the ‘Run Recipe’ button. From the recipe list, we used the recipe name ‘Source1-(E-beam) Cr process 100 nm’ and then pressed ‘Run’. ‘Source1-(E-beam) Cr process 50 nm’ was the recipe name for the 50 nm mask thickness. Then the system automatically managed the further steps and the progress of the e-beam deposition process was monitored through the Sigma software. The chromium source was heated with the electron emission to evaporate it. The system controls the temperature and power to get the suitable deposition rate of 3 Å/s. Then the shutter of the sample holder was opened automatically to start the deposition process. At the same time, the sample holder was rotated with the speed of 20 rpm to have uniform coating all over the substrates surface. When ‘Recipe complete’ appeared, ‘Ok’ was selected and the high voltage source was turned off. Then the chamber venting was performed and before unloading the wafer all the metal parts inside the chamber were touched with the grounding rod to avoid any possible electric shock. A dummy mask plate was placed on the sample holder to prevent any deposition to the heating lamps. Finally, ‘The PC Pump’ command was performed again.

3.1.3 Resist Coating

AR-P 6200 positive e-beam resist was used and coated with a Headway spinner PWM101D. The resist concentration of 1:1 was used during the fabrication of 50 nm and 2:1 during the fabrication of 100 nm mask thickness. First, the wafer was cleaned by the Nitrogen blow and placed on center of the rotating wheel of the spinner. The spinner was turned on temporarily to adjust the rotating speed of 2000 rpm and 1500 rpm during the fabrication of 50 nm and 100 nm mask thicknesses, respectively. The spinning time was set to 60 seconds for both of these mask thicknesses. 2 ml resist was taken using a pipette and carefully released onto the midpoint of the wafer surface. At this point, it was assured that no bubbles form/remain in the

resist. Immediately, the spinner was turned on and it stopped automatically after 60 seconds. Then the wafer was placed carefully on an preheated hot plate (HP atv) of 150° C temperature in order to evaporate the remaining resist solvent for 3 minutes. Finally, the spinner chamber and rotating stage were cleaned with acetone.

3.1.4 Electron Beam Patterning

The electron beam patterning was carried out using a Vistec EBG5000+ES HR system. A negative of the desired pattern was designed by AutoCAD software with desired grating parameters and saved as a DXF file format. Then the file was converted to a GPF file format using Beamer software with the writing grid resolution of 10 nm, step size of 20 nm and main field of 200 $\mu\text{m} \times 200 \mu\text{m}$. Note that, during designing of the grating negative on the AutoCAD, the grating groove width was reduced by 20 nm from the both sides which were then compensated by the dose level. Also the grooves width should be multiple of the resolution that we had used during creating the GPF file on the Beamer software. The GPF file was then transferred onto the Vistec EBG5000+ES HR system and using Cjob software the dose of 210 $\mu\text{C}/\text{cm}^2$ and 190 $\mu\text{C}/\text{cm}^2$ were set for the grating periods of 200 nm and 500 nm, respectively. The size of the grating was 5 mm $\times$ 100 μm and 5 mm $\times$ 225 μm for the grating period of 200 nm and 500 nm, respectively. The exposure time was around 58 minutes for these parameters.

Finally, machine readable exposure script JOB file format was generated using the Cjob software. The wafer was then placed on the sample holder and after venting the system it was loaded onto the chamber. The system pumps down to vacuum level of 5×10^{-5} mbars when the patterning was initiated. A basic workflow of a typical EBL system operation is illustrated in Fig. 3.2.

3.1.5 Resist Development

The exposed resist was developed with the help of OPTIspin SST20 automatic resist development system. The developer solution was EEP (Ethyl 3-Ethoxypropionate). The sample was placed on the spinning holder of the system and then the compartment door was closed. The N₂ gas line was opened and choose Return>Select> the recipe name 'EEP60S-100MM'>Automatic>Initialize from the control window. After system initialization, the 'Start' button was selected to run the development process. The system then released the developer solution onto the wafer surface to

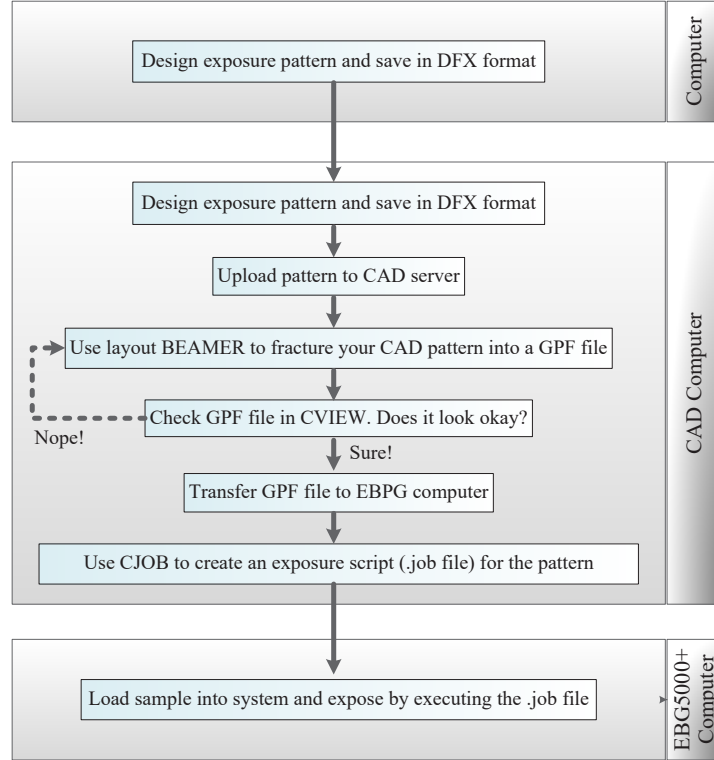


Figure 3.2: Basic workflow of a typical EBL system [48].

develop the resist for 1 minute. Then the system washed the wafer with isopropanol for 30 seconds followed by rinsing process with water. After that, the wafer was dried by spinning and then we unloaded it and closed the gas line. Finally, the wafer was blew by a Nitrogen gun to dry out the remaining water.

3.1.6 Dry Etching of Chromium

An Oxford Instrument Plasmalab 100 plus system was used to dry etch the chromium. In this process the ICP–RIE process was employed. First, we checked all the required gas lines are in open position. Before initiating the etching process, the etching chamber was cleaned with oxygen plasma cleaning for 20 minutes to remove any existing residue. The process parameters were: O_2 gas flow rate of 30 sccm, chamber pressure of 60 mTorr, temperature of 40° C and RF power of 300 W. Then the chamber was vented and a small amount of vacuum oil was used on the sample

holder in order to improve the thermal conductivity with the sample. After loading the sample, the oxygen plasma cleaning was carried out again for 7 seconds. The process parameters were: O₂ gas flow rate of 30 sccm, chamber pressure of 60 mTorr, temperature of 40° C, RF power of 100 W. Then the chromium etching was performed with the Cl₂/O₂ gas flow rate of 54/4 sccm, chamber pressure of 15 mTorr, temperature of 40° C, ICP power of 1500 W and RF power of 15 W. The DC Bias voltage of above 100 V was achieved for every wafer which is very important as it should be over 100 V to etch chromium properly. The etching process was carried out for 5:30 minutes and 2:30 minutes for the 100 nm and 50 nm chromium masks, respectively. After chromium etching, the oxygen plasma cleaning was performed again for 2 minutes to remove the remaining resist layer. The process parameters were the same as mentioned earlier. Then the system vent command was initiated to unload the wafer. Finally, backside of the wafer and sample holder were wiped with isopropanol to remove the vacuum oil.

3.1.7 Wafer Dicing and Cleaning

An automatic dicing saw DAD3240 was used to cut the wafer accordingly to have the individual 16 samples. After cutting the wafer, all the pieces were washed first in acetone and then in isopropanol for 1 minute in each chemicals. The pieces were then stored in the sample container and they were ready for the etching experiment. It is worth to mention that from each wafer we used one sample to investigate the mask thickness and its quality before performing the etching experiment.

3.2 ICP Etching of SiO₂

An Oxford Instruments Plasmalab 80 plus system with the ICP65 unit was used to study and develop the ICP etching process of SiO₂. The ICP unit had 65 mm source diameter and the ability of maximum 600 W RF power with automatic vacuum capacitor matching. The electrode of the system was powered by a 13.56 MHz RF generator. The system was connected to the SF₆, CF₄, CHF₃, O₂, N₂ and Ar gases as plasma sources and He as a backside substrate coolant. Some other controllable parameters of the system were RF power, ICP power, chamber pressure, chiller temperature, strike rate and ramp rate.

A picture of such a system is shown in Fig. 3.3 with the chamber controllers. The

system venting and pumping were managed in the pumping control page which was obtained by selecting Menu>Pumping on the system computer. The system venting was initiated by selecting ‘STOP’ and then the ‘VENT’ button. It required few minutes to allow enough time for the turbo pumps to be purged. When ‘Vent Time Left’ decreased to zero, the chamber was opened by selecting ‘Up/Down’ selection switch to its up position and pressing both hoist buttons simultaneously. After loading the sample, the chamber was closed by selecting ‘Up/Down’ selection switch to its down position and again pressing both hoist buttons simultaneously. The chamber pumping was initiated from the pumping page by selecting ‘Stop’ and then ‘Evacuate’ button. A window was prompted by the system to write a sample name. A process/recipe was run from the Process>Recipe page. A previously saved recipe was loaded by selecting ‘Load’ button and then by selecting required recipe from the list and then ‘OK’ button. After loading the recipe, the process was executed by selecting ‘Run’ button. The recipe was edited or created by selecting on the recipe steps or blank Steps, respectively and then by selecting ‘Edit Step’. In that case, process step editor window opened. The corresponding process parameters were filled cautiously and then the ‘Run’ button was selected to execute the process. A process complete message was notified by the system at the end of each processes.

The ICP etching of the samples were carried out by changing different parameters of the etching process. The etching rate of the process was expected to be below 100 nm/min so that one can get a desired etching depth from nanometer to sub-micrometer scale. The investigated parameters were: RF power, ICP Power, chamber pressure, gas composition, flow rate of reactive gases, substrates backside cooling and etching time. The main reactive gases were SF_6 , CF_4 and CHF_3 . Each of these gases was combined with O_2 or Ar or both of them with different ratios during the study. The reactor chamber was cleaned by oxygen plasma cleaning for at least 30 minutes at the beginning of the experiment and before swapping between the reactive gases SF_6 , CF_4 and CHF_3 . The oxygen plasma cleaning parameters were: O_2 gas flow rate of 20 sccm, RF power of 300 W, chamber pressure of 60 mTorr, and Strike pressure of 80 mTorr. When He-backside cooling was applied, we used vacuum oil on the sample stage in order to improve the thermal conductivity with the sample.

At earlier stage of the study, we roughly found out a suitable selectivity by investigating different RF powers of the etching process. Once we had the suitable

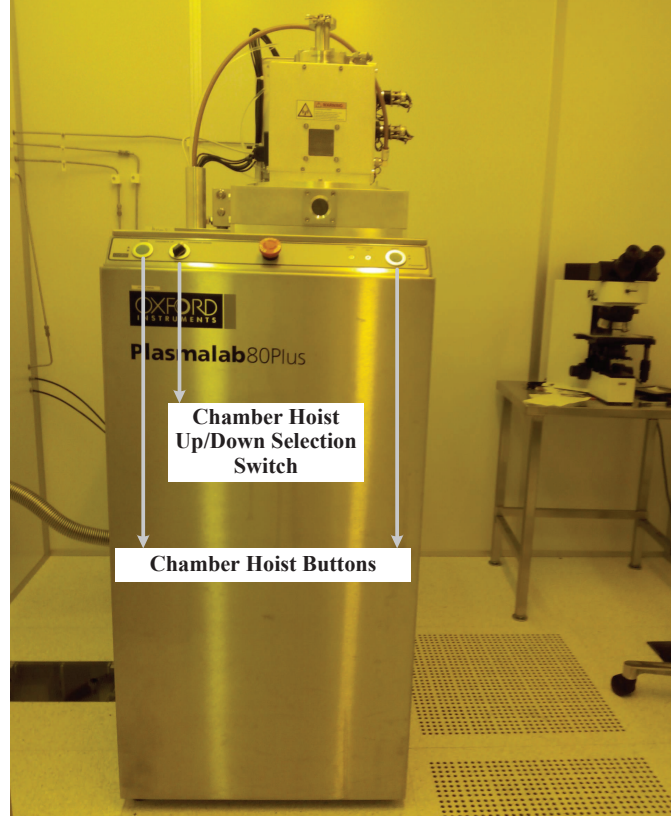


Figure 3.3: The Plasmalab 80 plus system with the ICP65 unit.

selectivity and its corresponding RF power we studied the effects of RF power, ICP power, chamber pressure, different gas compositions and the effect of He-backside cooling. By analyzing and comparing all of these studies and corresponding profiles quality we have solved the optimized parameters. Sidewall angle, flatness, roughness, polymer deposition, some etching effects, like micro-trenching, bowing, loading effect, RIE lag effect, etc. were considered during the profile quality analysis. Furthermore, the process repeatability and the change of fill factor of the grating with different etching times were investigated for the optimized process. The detailed procedure of the selecting optimized parameters is described in the following Chapter.

3.3 Characterization

Each samples was characterized through the help of SEM. After the SiO_2 etching with ICP, the sample was cut down through the gratings manually using a glass cutter pen to investigate the cross-section of the gratings. Using the pen a sample was scratched on the both side of two consecutive gratings. It was then twisted against the sketched line to split it. One of the fragments was then used for SEM imaging.

3.3.1 Sputter Coating

Coper or Gold was coated over the cross-section of the gratings to make a conductive layer for SEM imaging. The coating was done using an Emitech K675X sputter. Argon gas was used to form plasma in the system. These plasma ions then strikes the source to defuse and consequently coated onto the sample. The terminate value of the process, that is, the thickness of the coated layer was 10 nm.

3.3.2 Scanning Electron Microscope Imaging

A SEM Leo 1550 Gemini system was used to investigate the grating profiles. First, the chamber venting was performed from ‘Vacuum’ tab by selecting Vent in the system control software. The sample was always mounted in a sample holder that can hold the sample vertically to take cross-section images. After venting the system, the sample holder was loaded in the chamber and the door was closed. The pump down process was initiated by selecting ‘Pump’ in the ‘Vacuum’ tab. When the ‘Vacuum Status’ was in ‘Ready’ state, we switched ‘EHT’ to ‘On’ from the ‘Gun’ tab. Through the system control panel, the sample was positioned just under the lens by adjusting the position knob. Then ‘Camera’ was selected to switch the ordinary camera mode to the SEM mode. The scanning speed was adjusted to have a clear image. Then we found out the grating position on the sample using low magnification. After finding the grating position, the magnification was increased to 50000X. At low scanning rate the stigmations and focus were adjusted simultaneously to have a clear image. The grating parameters were measured or the image was taken by freezing the live scan. Since SEM measurements at different magnifications shows different values, all measurements, like mask thickness, grating height, groove width, etc. were carried out using the same magnification of 50000X. Some images were taken at the higher magnification to investigate the profile quality, like sidewall surface roughness.

Results and Discussions

In this Chapter, we will present, analyse and compare the deferent etching outcomes of the both gratings for different etching parameters and conditions. The change of etching rate, selectivity and DC bias were investigated only for the grating period of 500 nm and presented here as a function of different etching parameters for comparison. However, in terms of profile quality analysis, both grating periods were considered. Sidewall angle, flatness, surface roughness and polymer deposition were investigated and compared to find out the possible best profile and its corresponding recipe. Some etching effects, like micro-trenching, bowing and loading effect were also considered in the evaluation. The etched outcomes for two different mask thicknesses of 50 nm and 100 nm were also analysed and compared. Finally, the change of fill factor and etching rate with different etching times and repeatability of the possible best process were investigated and presented here.

From all of the etching parameters and conditions, the etching gas composition was found to have very significant role in the etched profile structure. Different combination of gases were used to etch SiO_2 during the study. The main reactive gases were SF_6 , CF_4 and CHF_3 . Each of these gases was combined with O_2 or Ar or both of them with different ratios during the study. At the initial phases of the study, it was observed that the profile quality and selectivity were poor for higher RF power regardless of any other parameters except ICP power, especially selectivity was very poor above 50 W RF power with ICP power of 600 W. Profile quality and selectivity were also poor for chamber pressure above 15 mTorr with the same ICP power. So, later on, mostly we have investigated rest of the samples with bellow RF power and chamber pressure of 50 W and 15 mTorr, respectively. Unfortunately

in our system, the etching processes were mostly unstable with the CF_4 reactive gas at the below RF power and pressure of 50 W and 15 mTorr, respectively. In this case, we have observed the fluctuation of reflected power and RF power which causes the fluctuation of DC bias arbitrarily. As a result, the process was non repeatable and we did not get any consistent etching rate, selectivity and profile structure for investigation. Hence, the investigation with the CF_4 reactive gas was limited by this issue and its etching results were ignored in this study. We have observed similar issue with other reactive gases only for certain cases which were also ignored. The cause of these unstable etching processes are unknown and requires further investigation. It is worth to mention that the results of the etching rate and selectivity are not absolute. These results were measured with the help of SEM to get rough estimation and to compare with the different etching outcomes. Few parameters, like pressure strike, ramp rate and chiller temperature were kept fixed during the investigation which were 60 mTorr, 5 mTorr and 20°C , respectively.

4.1 Influence of RF Power

At the beginning of the study, several range of RF power were applied to investigate the selectivity. As mentioned earlier, with 600 W ICP power, selectivity and profile quality were poor above 50 W RF power and better at the range of 20 W – 30 W RF power for all process gases. Hence we mostly investigated the etching of SiO_2 within this range of RF power when 600 W of ICP power was applied. However, one can also get similar selectivity with the higher RF power by introducing He-backside cooling with the process parameters. We will discuss this case in Section 4.7. Fig. 4.1 and Fig. 4.2 represent the change of etching rate and DC bias as a function of RF power with the CHF_3/Ar and SF_6/Ar plasma, respectively. Fig. 4.3 represents the change of selectivity as a function of RF power. The process parameters were: CHF_3/Ar or SF_6/Ar gas flow rate of 10/15 sccm, ICP power of 600W, chamber pressure of 9 mTorr, and etching time of 10 minutes. As expected, the DC bias and etching rate increase almost linearly with increasing RF power within this selected range for both CHF_3/Ar and SF_6/Ar plasma. This result was expected because the plasma particles gain more energy with the higher RF power which then can bombard harshly onto the sample surface and increase the amount of physical etching. The selectivity was found to be decreased with increased RF power. The mask material

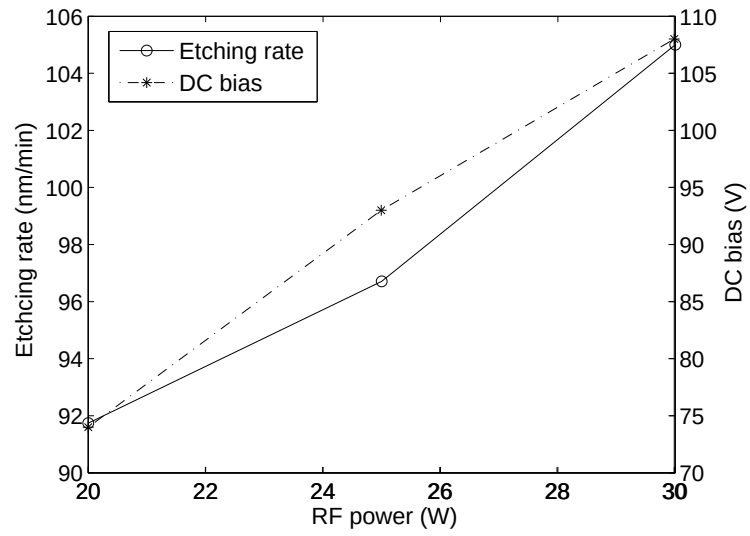


Figure 4.1: Etch rate and DC bias as a function of RF power for the CHF_3/Ar plasma.

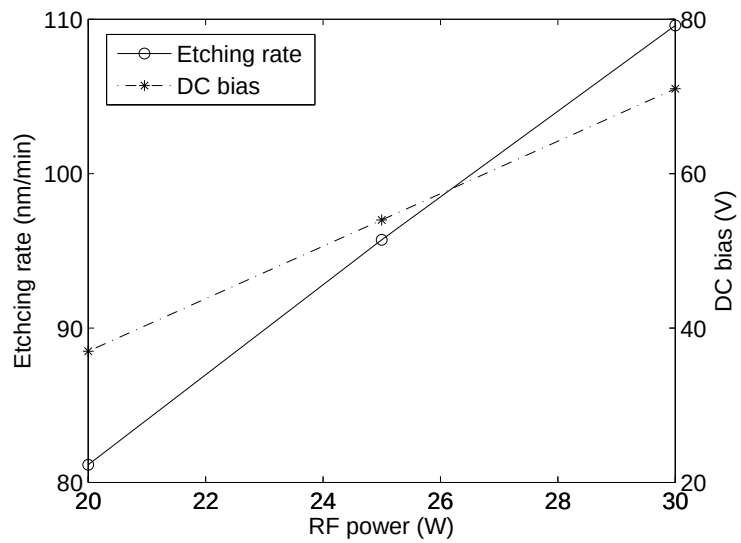


Figure 4.2: Etch rate and DC bias as a function of RF power for the SF_6/Ar plasma.

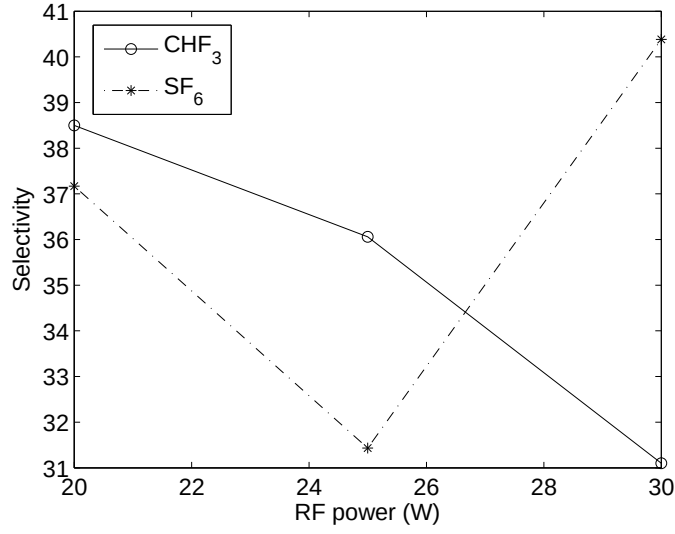


Figure 4.3: Selectivity as a function of RF power.

chromium is mostly etched by the physical etching rather than the chemical etching. Since the physical etching is more significant with the higher RF power which causes the chromium layer to etch faster. This leads to the poor selectivity at the higher RF power range. We also investigated the etching characteristics by changing other etching parameters near this range of RF power. It was found that, in most cases polymer deposition dominates below the RF power of 20 W for the CHF₃ and 15 W for the SF₆ reactive gases. Profile quality is clearly better near the 20 W RF power even for the different gas compositions and chamber pressures with the SF₆ reactive gas. However, for the CHF₃ reactive gas, the profile quality varies depending onto the gas composition and chamber pressure within the range of 20 W – 30 W RF power. Undercutting was minimum for the RF power of 20 W compared with the 30 W. Hence, further investigations were carried out by keeping these factors in mind.

4.2 Influence of ICP Power

The etching characteristics was investigated with the ICP power of 200 W, 400 W and 600 W. Figs. 4.4 and 4.5 represent the change of etching rate and DC bias as a function of ICP power for the CHF₃/Ar and SF₆/Ar plasma, respectively. The change of selectivity as a function of ICP power is shown in Fig. 4.6. The process

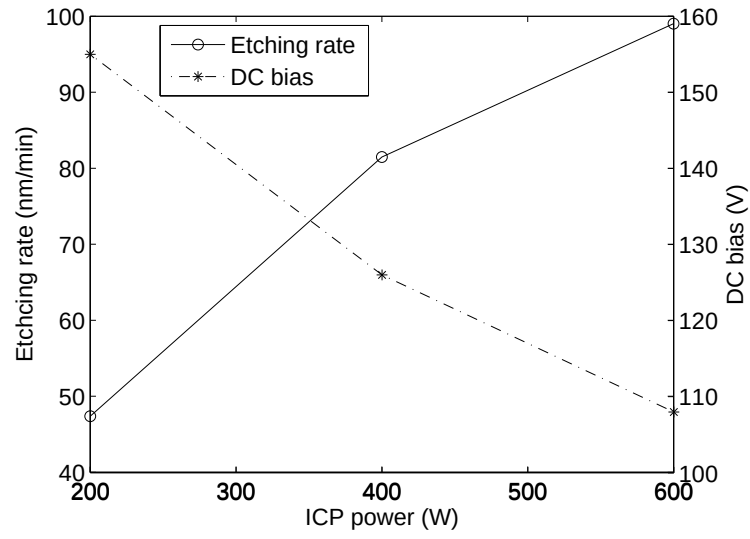


Figure 4.4: Etch rate and DC bias as a function of ICP power for the CHF_3/Ar plasma.

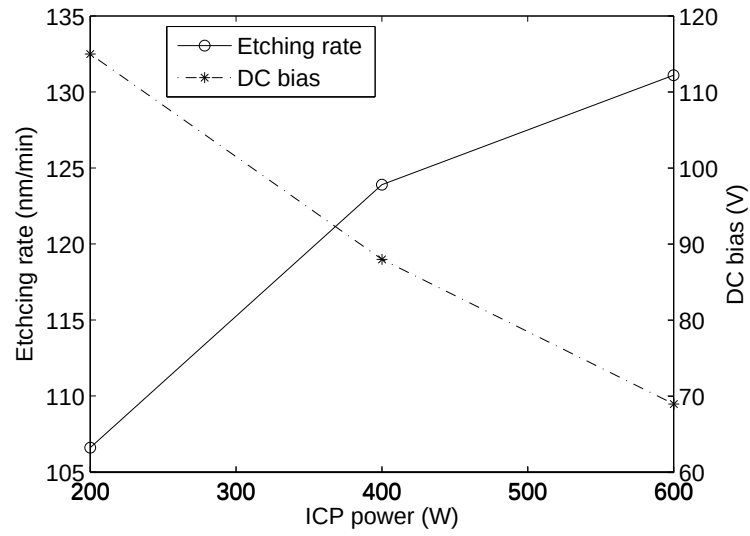


Figure 4.5: Etch rate and DC bias as a function of ICP power for the SF_6/Ar plasma.

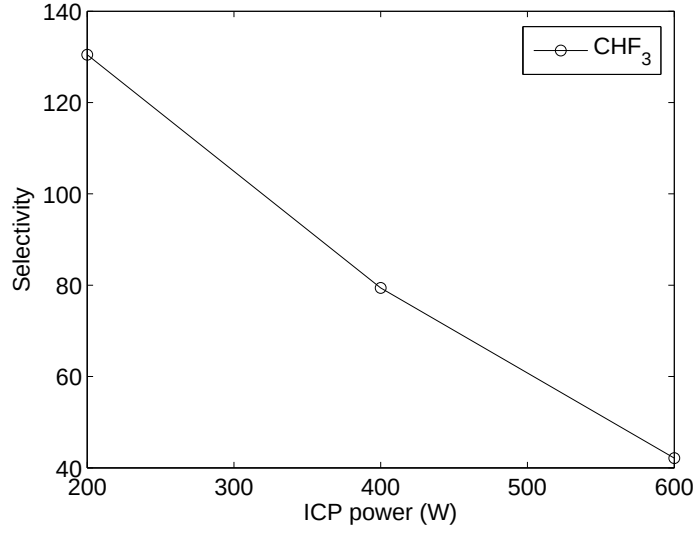


Figure 4.6: Selectivity as a function of ICP power for the CHF_3/Ar plasma.

parameters were: CHF_3/Ar or SF_6/Ar gas flow rate of 10/15 sccm, RF power of 30 W, chamber pressure of 9 mTorr, and etching time of 10 minutes. The strike pressure was 90 mTorr in the case of 200 W ICP power for the SF_6/Ar in order to start the plasma. In other cases it was the default value of 60 mTorr. It was found that the DC bias start to increase as the ICP power decreases. As expected, etching rate increases with increasing ICP power, since the ion density of the plasma in the chamber increases with the increased ICP power. However, too high ICP power could lead to downward trend in the etching rate due to the mean free path of the plasma molecules gets shorter with the excessive ICP power [15]. As a result, collision among plasma particles increases and loses there energy before strike onto the substrate. For the CHF_3 reactive gas, etching rate drops dramatically compared with the SF_6 reactive gas due to the enhanced polymer deposition at the lower ICP power of 200 W. As a result we have seen too much high selectivity at the lower ICP power of 200 W. However, excessive polymer deposition is not desirable and one can get rid of it by increase the RF power. The etching rate for the SF_6/Ar plasma was higher and above 100 nm/min even with the ICP power of 200 W. For this reason, the selectivity estimation was critical with the 50 nm mask and 10 minutes of etching time. However, roughly the trend of selectivity was remained flat with the change of ICP power. By considering overall profile quality, like sidewall angle,

micro-trenching, polymer deposition, profile structure, etc. we had continued our investigation with the ICP power of 600 W. However, there is plenty of opportunity to investigate below 600 W ICP power and one might be able to get better optimized parameters.

4.3 Influence of Pressure

One of the most important benefits of using ICP that it has the ability to operate at low pressure with high plasma density. We have investigated the etching process in the range of 3 mTorr – 15 mTorr pressure. Fig. 4.7 represents the change of etching rate as a function of chamber pressure for the both reactive gases. The RF power was 30 W and 20 W for the CHF₃/Ar and the SF₆/Ar plasma, respectively. Other parameters were CHF₃/Ar or SF₆/Ar gas flow rate of 10/15 sccm, ICP power of 600 W and etching time of 10 minutes.

From the figure, it is clear that the etching rate is dropping towards the higher pressure. This is understandable, since the higher chamber pressure increases the particles density, lower the mean free path, increases the collisions and hence, reduces the plasma particles energy. The selectivity also drops towards the higher pressure especially for the CHF₃/Ar compared with the SF₆/Ar plasma. The fill factor of the grating also changes towards the higher chamber pressure, specially for the CHF₃/Ar plasma. The groove width starts to increase towards the higher pressure significantly. This was expected, since we have already discussed in the theory that higher chamber pressure reduces the directional etching. Polymer deposition takes place on the uppermost part of the sidewall at the lower chamber pressure range of 3 mTorr – 9 mTorr. It is worth to mention that this kind of deposition was found to be higher in the sample having mask of 100 nm. For comparison, two SEM images are shown in Fig. 4.8, etched with the chamber pressure of 3 mTorr and 15 mTorr for the CHF₃/Ar plasma. The process parameters were the same as mentioned earlier. However, in the case of SF₆/Ar plasma the structure does not change noticeably over the selected pressure range and no significant polymer deposition was found. Note that, in some cases we have found a few unstable processes with the SF₆/Ar plasma for different gas compositions at the chamber pressure of 3 mTorr and we were forced to use the 9 mTorr of chamber pressure for investigation further.

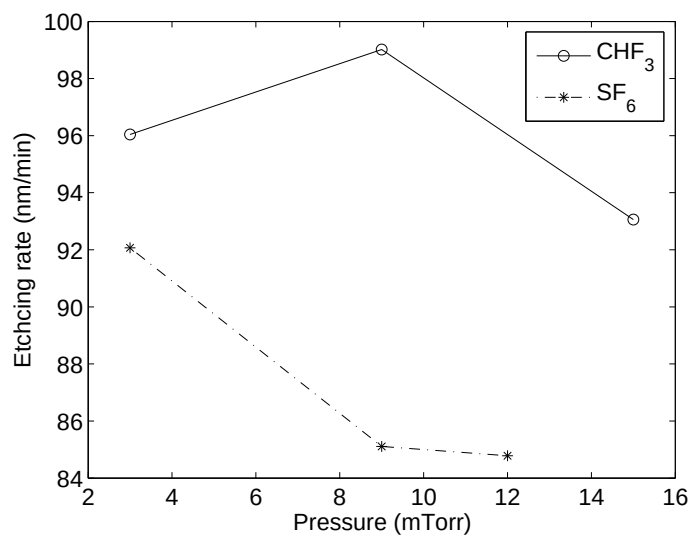


Figure 4.7: Etch rate as a function of pressure.

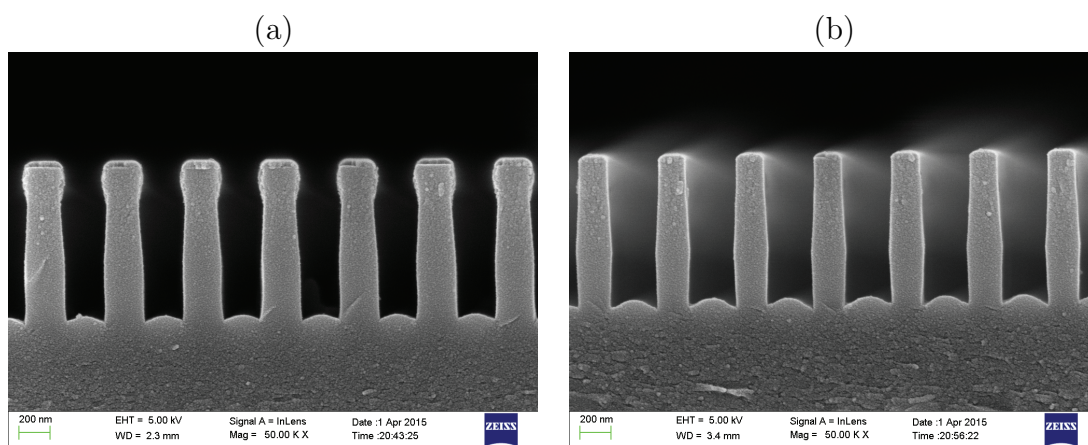


Figure 4.8: Profiles of the grating having period of 500 nm etched with the CHF₃/Ar plasma at the chamber pressure of 3 mTorr (a) and 15 mTorr (b).

4.4 Effect of Ar Addition

As we discussed in the theory, inert gases, like He or Ar can be used as an additive gas within fluorine-containing plasma. The etching characteristics was investigated for different combinations of Ar addition. By keeping the CHF_3 or SF_6 flow rate fixed, we have increased the Ar flow rate from 33% to 75% of the total gas flow (including Ar). The etching parameters were: CHF_3 or SF_6 gas flow rate of 10 sccm, RF power of 30 W, ICP power of 600 W chamber pressure of 9 mTorr and etching time of 5 minutes. As we can see from the Fig. 4.9 the etching rate is quite constant up to 50 % of Ar for the both CHF_3 and SF_6 reactive gases. However, it starts to drop beyond that due to the excessive Ar ion density, particularly in the case of CHF_3 reactive gas. In this case, the electron density remains almost same but the average electron energy decreases which reduces the etching rate [16]. Overall profile structure remains almost the same over the selected range of Ar addition. Particularly in the case of CHF_3 reactive gas, the polymer deposition increases slightly for more than 60% of Ar addition. Another but most important observation was that the least micro-trenching at 60% of Ar addition compared with the other percentages of Ar. Similar observation was reported earlier in the case of fluorine-containing plasma etching [16]. Consequently we have selected this 60% of Ar addition as an optimal parameter.

4.5 Effect of O_2 Addition

O_2 addition in plasma gases plays very significant role in the etching profiles quality. We have investigated the effect of 1 sccm and 2 sccm O_2 addition with the different gas compositions, RF power, chamber pressure and also with the He-backside cooling. In general, etching rate increases with the O_2 addition. In the case of CHF_3 reactive gas etching rate increases about 5 to 10% depending onto the other parameters. However, in the case of SF_6 reactive gas it increases to some extent only. We never observed any significant change of the selectivity due to the 1 sccm of O_2 addition for both reactive gases. However, the profile quality is very sensitive to the addition of O_2 . We have seen distinct undercutting for only 1 sccm O_2 addition with the CHF_3/Ar plasma. More undercutting was observed without the presence of Ar gas. It also reduced the micro-trenching and the polymer deposition on the sidewall for the CHF_3/Ar plasma. Fig. 4.10 represents two SEM images to compare the profiles that etched with and without O_2 addition with the CHF_3/Ar plasma.

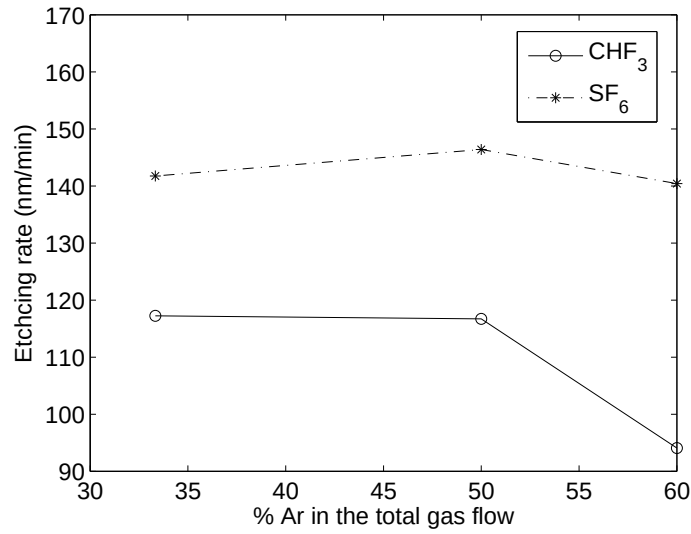


Figure 4.9: Etch rate as a function of Ar percentage.

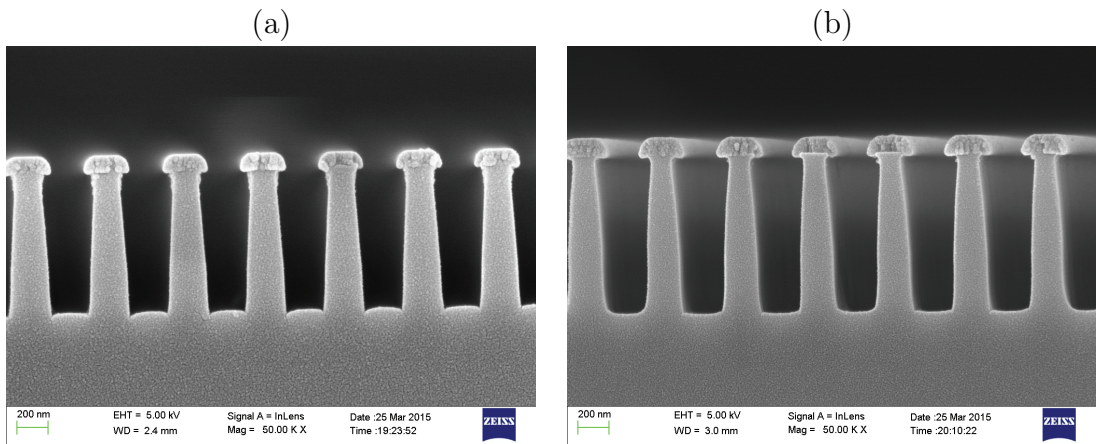


Figure 4.10: Profiles of the grating having period of 500 nm etched without addition of O₂ (a) and with addition of 1 sccm O₂ with the CHF₃/Ar plasma (b). The process parameters were: CHF₃/Ar gas flow rate of 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 9 mTorr and etching time of 10 minutes.

However, this is not the case for the SF_6 reactive gas. As we can see from the Fig. 4.11, the undercutting and polymer deposition under the mask remain about the same but the micro-trenching increases significantly with the addition of O_2 gas. Therefore, we recommend not to add O_2 gas with the SF_6 reactive gas in order to get least micro-trenching in the profile.

4.6 Influence of Total Gas Flow Rate

The effect of different gas flow rates have been studied for the both SF_6 and CHF_3 reactive gases. However, the change of etching rate as a function of total gas flow rate is shown in Fig. 4.12 only for the CHF_3/Ar plasma. The etching parameters were: Ar to CHF_3 ratio of 1.5 (60% Ar), RF power of 20 W, ICP power of 600 W, chamber pressure of 3 mTorr and etching time of 10 minutes. As expected, the etching rate increases with the increased flow rate over the selected parameters. With the total gas flow rate of 12.5 sccm, the sidewall of the profile looks more vertical compared with others amounts but the fill factor of the grating changes rapidly with the different etching times for this amount of flow rate. For the flow rate of 37.5 sccm, we have observed more slanted sidewall compared to the profile achieved with the 25 sccm of flow rate. In the case of He-backside cooling with the 37.5 sccm of flow rate, the polymer deposition becomes dominant and the etching process stops after few 100 nm of etching. For these reasons we have selected the 25 sccm of total gas flow rate as an optimal parameter.

4.7 Influence of He-backside Cooling and Sidewall Passivation Analysis

The reactive species produce unsaturated compounds in the plasma, leading to polymer formation and deposition on the substrate. Proper amount of polymer formation is required on the profiles sidewall to protect it from being etched by the plasma. One might be able to control it by controlling the temperature of the etched sample. Helium-backside cooling was used to keep the sample cool during the etching process. Typically it reduces the etching rate for any plasma gases. But the effect is substantial in terms of the sidewall passivation.

Influence of the He-backside cooling was investigated by applying 5 Torr and

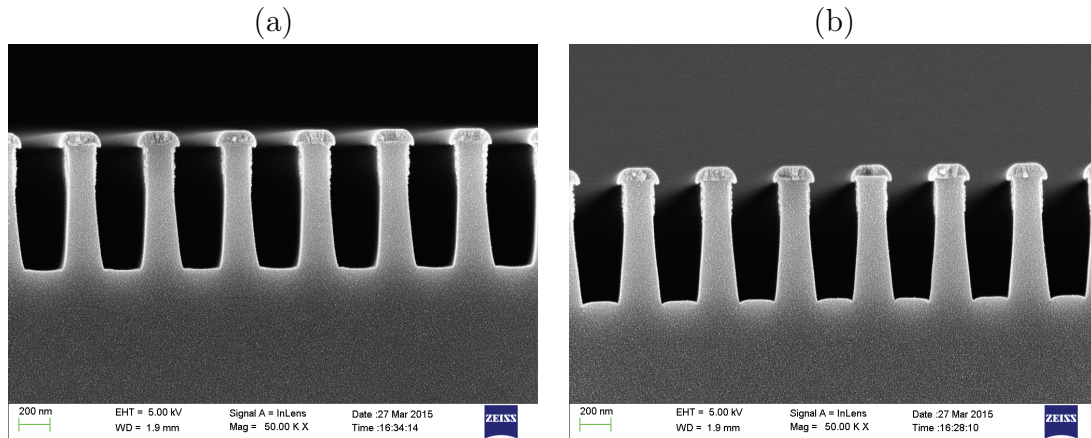


Figure 4.11: Profiles of the grating having period of 500 nm etched without addition of O_2 (a) and with addition of 1 sccm O_2 with the SF_6/Ar plasma (b). The process parameters were: SF_6/Ar gas flow rate of 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 9 mTorr and etching time of 10 minutes.

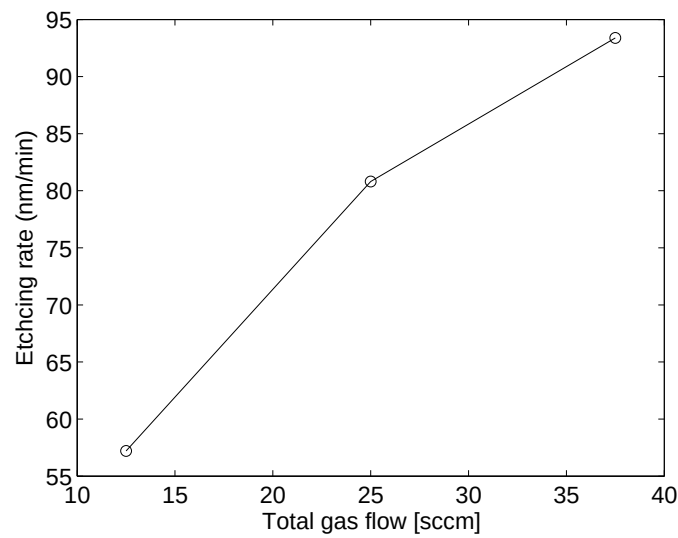


Figure 4.12: Etch rate as a function of total gas flow rate for the CHF_3/Ar plasma.

10 Torr of He pressure with the different etching parameters. For the SF_6/Ar plasma, the etching rate drops more than 10% for all cases and excessive polymer deposition occurred especially at the RF power below 30 W due to the He-backside cooling. It was then reduced by increasing the RF power to 40 W but in this case, we have observed slanted and wavy sidewall compared to the other etching outcomes. So we do not recommend to apply the He-backside cooling in the case of SF_6/Ar plasma.

However, the profile quality of both gratings improved significantly with the He-backside cooling for the CHF_3/Ar plasma. We were able to achieve nearly 90° (for the grating period of 500 nm) and comparatively flat sidewall by applying the 5 Torr of He pressure with the CHF_3/Ar flow rate of 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 3 mTorr and etching time of 10 minutes. The etching rate of the recipe was 72.8 nm/min which is about 3.2% less than the same recipe without the He-backside cooling. The SEM images of the both grating periods of 200 nm and 500 nm are shown in Fig. 4.13 (a) and (b), respectively for the above mentioned etching parameters. However, the great failure of the process was that the etching of SiO_2 stops at some point for more than the 10 minutes of etching time. As we can see in Fig. 4.14, thick layer of polymer deposited onto the sidewall and etched product remained on the bottom of the grooves for the 15 minutes of etching time. It was observed for the CHF_3/Ar plasma that the DC bias starts to fall slowly after about 8 minutes of etching time until it drops about 10 V depending onto the total etching time. This phenomenon was very found very common in our ICP system with the CHF_3/Ar plasma. As we have mentioned in Section 4.1, the polymer deposition started to increase below the RF power of 20 W or in other words below the corresponding DC bias, that is, below the DC bias on average of 75 V. Besides this issue, the He-backside cooling just enhanced the process of polymer deposition. Thus, we have increased the RF power to 30 W to solve this problem. The technique worked well but as we can see in Fig. 4.15, it results a thin layer of polymer that deposited only on the uppermost part of the sidewall and the profile looks like it has weak bowing effect.

On the other hand, we have observed thick layer of polymer deposition on the sidewall with 10 Torr of He pressure for the CHF_3/Ar plasma. The effect was checked also with the higher RF power of 40 W. Hence, we do not recommend to apply 10 Torr of He pressure for the CHF_3/Ar plasma.

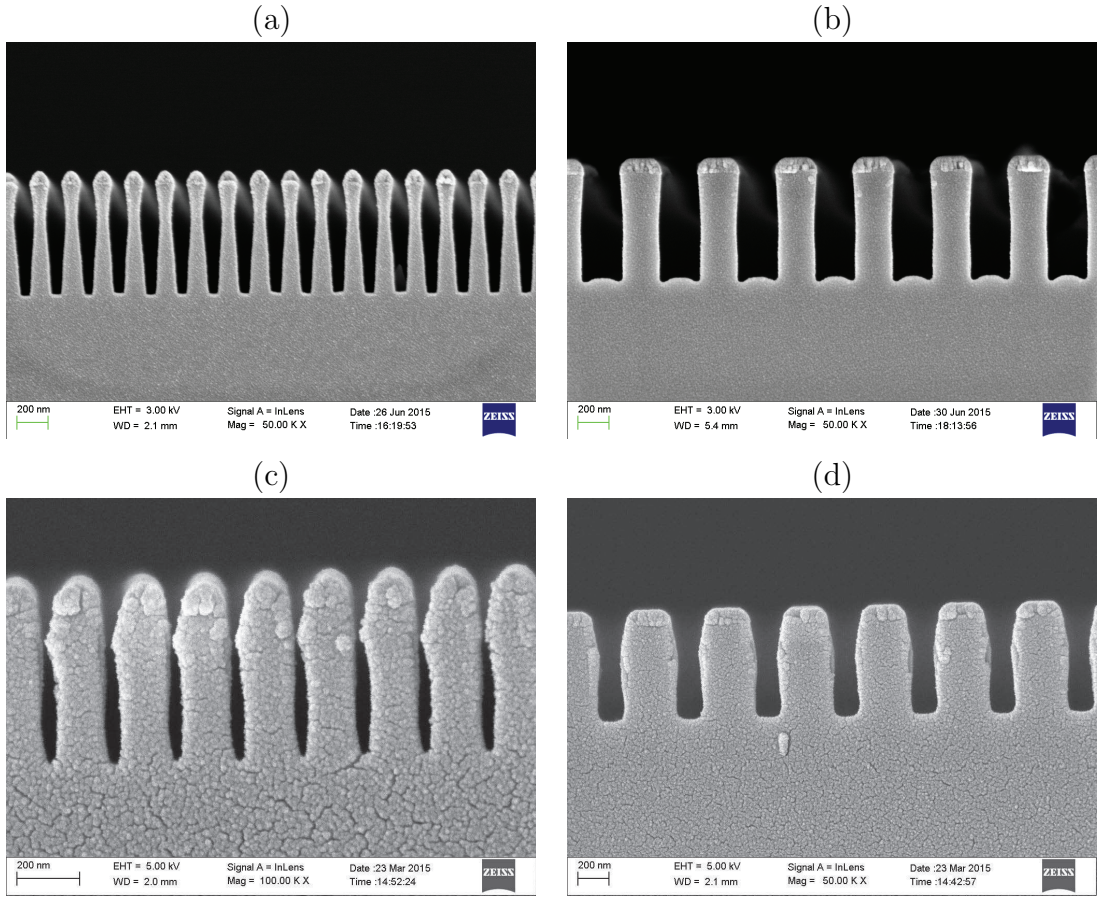


Figure 4.13: Cross-sectional SEM images of the gratings having periods of 200 nm (a) and 500 nm (b) etched with the gas flow rate of CHF_3/Ar 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 3 mTorr, He pressure of 5 Torr (backside cooling) and etching time of 10 minutes. The grating height was 722 nm and 718 nm for the periods of 200 nm and 500 nm, respectively. While, two SEM images of the grating having periods of 200 nm (c) and 500 nm (d) were etched with the RIE for comparison. The parameters were: CHF_3/Ar gas flow rate of 12/38 sccm, RF power of 220 W, chamber pressure of 45 mTorr, etching rate of 38 nm/min, and etching time of 15 minutes. The grating height was 568 nm and 639 nm for the periods of 200 nm and 500 nm, respectively.

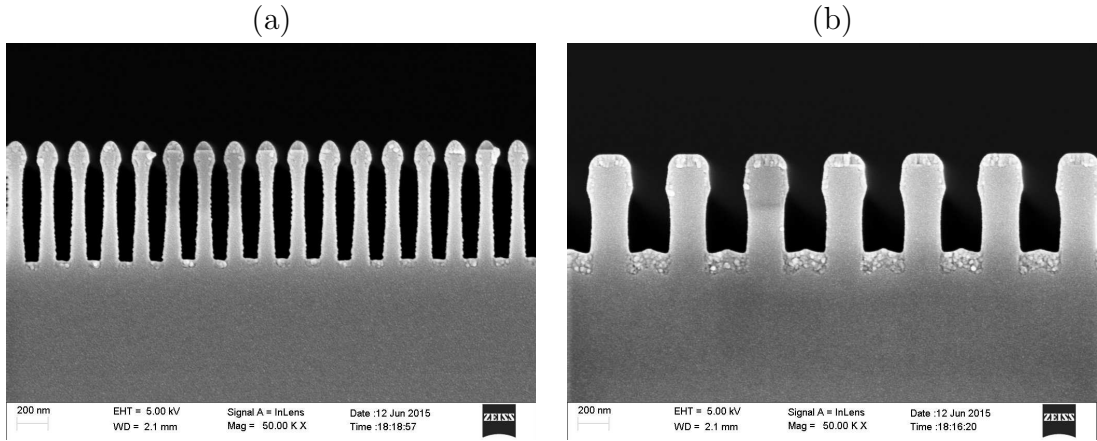


Figure 4.14: Cross-sectional SEM images of the gratings having periods of 200 nm (a) and 500 nm (b) etched with the gas flow rate of CHF_3/Ar 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 3 mTorr, He pressure of 5 Torr (backside cooling) and etching time of 15 minutes.

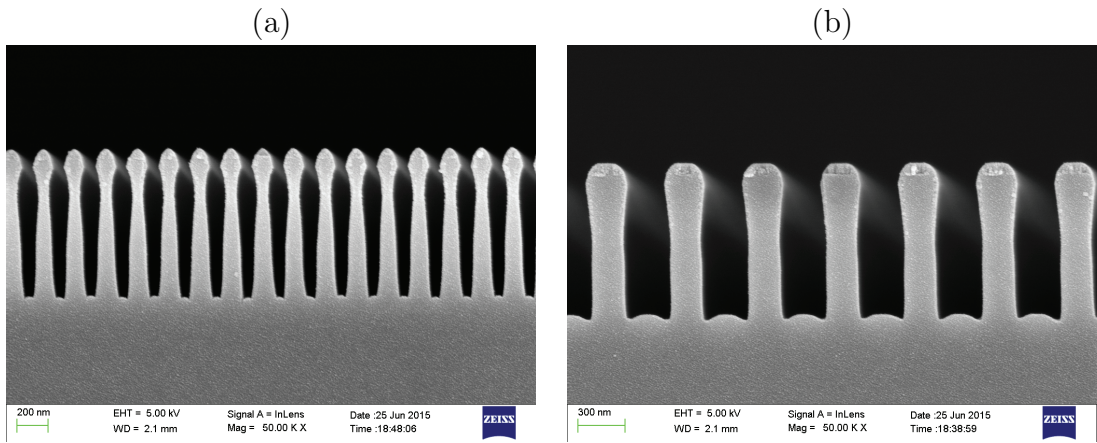


Figure 4.15: Cross-sectional SEM images of the gratings having periods of 200 nm (a) and 500 nm (b) etched with the gas flow rate of CHF_3/Ar 10/15 sccm, RF power of 30 W, ICP power of 600 W, chamber pressure of 3 mTorr, 5 Torr of He pressure (backside cooling) and etching time of 10 minutes. The etching rate was 90.1 nm/min and 94.1 nm/min for the periods of 200 nm and 500 nm, respectively.

4.8 Micro-trenching Analysis

The ICP tool is different from the conventional RIE reactor because it provides higher ratio of the ion flux to the reactive neutral flux onto the substrate. This condition can lead to micro-trenching [44]. We have faced this problem throughout the experiment and it was the most challenging factor to resolve. We have investigated the micro-trenching by changing the different gas compositions, RF power, ICP power, chamber pressure and by applying He-backside cooling. Nevertheless, it was found to be predominantly invariant. However, we have found some factors that affect the micro-trenching of the profile to some extent. As discussed in Section 4.4, the micro-trenching was minimum when 60% of Ar was used in the total gas flow. Oxygen also influences the charging effect and enhances the micro-trenching depending onto the plasma chemistry [49, 50]. Referring to Section 4.5, we also found this effect when SF_6/Ar plasma chemistry was used. Another remarkable observation was that the micro-trenching gets weaker with increasing etching depth in the case of SF_6/Ar plasma. However, this is not the case or rigorously, not strong enough for the CHF_3/Ar plasma probably due to the more charging effect than the SF_6/Ar plasma.

After thorough investigation, it was found that the sample which has better sidewall mask provides the least micro-trenching. This is understandable, since the initial development of the micro-trenching caused by the slope of the mask sidewall or influence of the electric field of the charged mask. We were able to achieve micro-trenching free profiles only for those samples correspond to the wafer having better masks sidewall. Quality of the mask was better for the 100 nm thickness than the 50 nm, presumably because of the higher concentration resist which was used during the mask fabrication process. Fig. 4.16 represents the best and worse mask profiles and their corresponding etched profiles for comparison. As we can see from the figure, not only the micro-trenching but also the sidewall quality of the etched profiles get effected with the mask quality. We have seen shallower groove width toward base of the groove due to the slanted sidewall of the mask. This particular case was verified also with other etching parameters. Hence we highly recommend to develop the mask fabrication process in order to get a profile with minimal micro-trenching and more vertical sidewall.

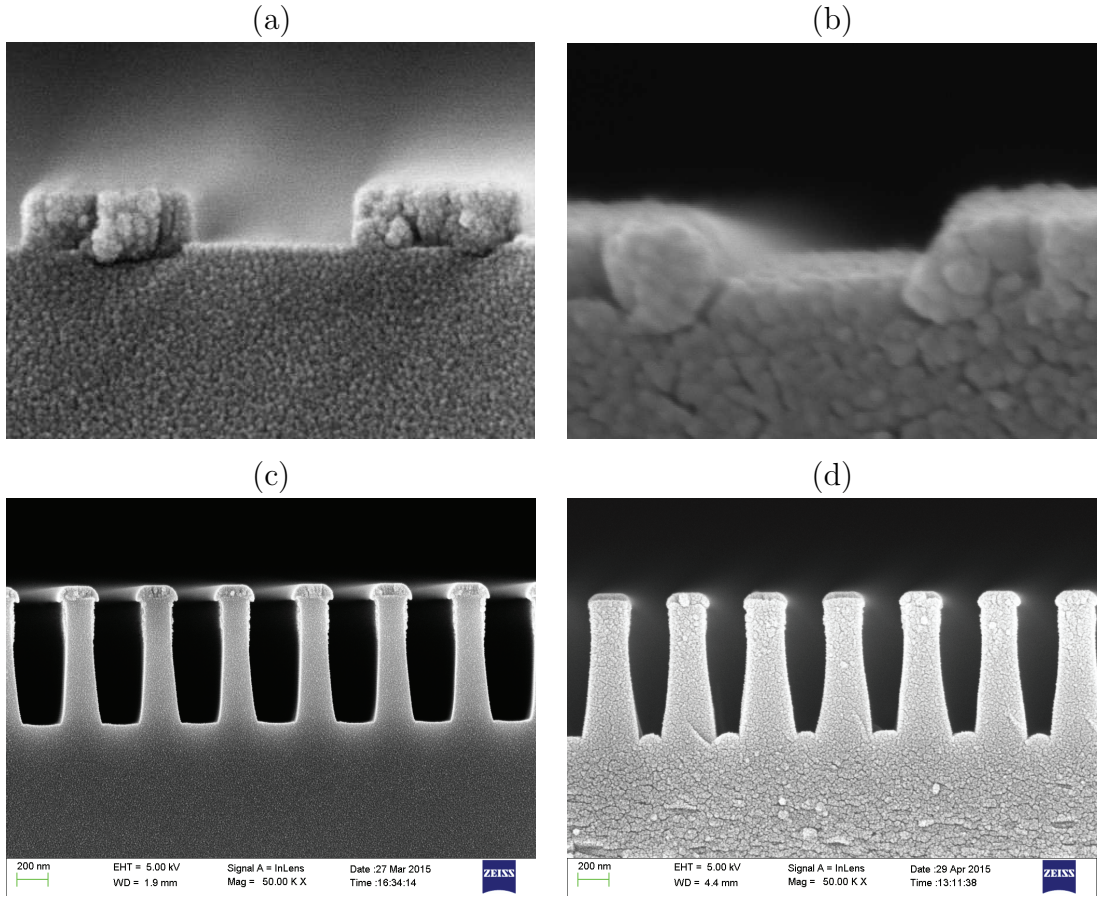


Figure 4.16: Profiles of the best (100 nm) (a) and the worse (50 nm)(b) wafer masks and its corresponding etched samples (c) and (d), respectively. The etching parameters were: SF_6/Ar gas flow rate of 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 9 mTorr and etching time of 10 minutes.

4.9 Determination of The Optimized Process

After numerous investigations it was realized that the optimisation of the ICP etching process is strongly dependent on the geometry of the structure. Different feature sizes require different optimized parameters to get the precise profile structure. However, one needs a typical recipe and characteristics of the process parameters to find out the best optimised parameters for the desired feature size. Hence we have investi-

gated and developed few recipes that could work for nanometer to micrometer scale.

In the earlier sections, we have discussed the step by step effects of the different process parameters. We have found the selectivity and profile structures are better at the range of RF power 20 W–30 W and the chamber pressure of 3 mTorr–9 mTorr with the ICP power of 600 W. The 60% of Ar addition to the total gas flow rate reduces the micro-trenching. Addition of the O₂ gas increases the undercutting, effects the micro-trenching and reduces the sidewall passivation. The total gas flow rate of 25 mTorr was found to be the best choice to investigate over the selected ranges of other parameters. In the case of CHF₃/Ar plasma, controlled amount of He-backside cooling improves significantly the sidewall passivation and provides flat and vertical sidewall. However, excessive backside cooling could lead to undesirable thick layer of polymer deposition on the sidewall. The grating having period of 200 nm had the better structure when it was etched with the CHF₃/Ar plasma compared with the SF₆/Ar plasma. The grating grooves and bars got triangular shape due to the slanted sidewall when the sample etched for the 10 minutes of etching time with the SF₆/Ar plasma. By considering all of these factors and as we have earlier mentioned that the different etching depths require different optimized parameters, we have solved four best optimised parameters which are presented in Table 4.1. Each of them has its own advantages and disadvantages. Note that, selectivity estimation was difficult throughout the experiment due to the different samples etched with the same process parameters showed different values, especially when the different etching times were used. However, typically it was well above 35 for all of these recipes.

From all of the mentioned recipes in the Table 4.1, recipe-1 is the best and provides finest profile in terms of vertical and flat sidewall for both gratings up to the 750 nm of grating height. Almost 90° and comparatively flat sidewall was achieved for the grating period of 500 nm. However, it has some limitations which are already discussed in Section 4.7. We have checked the repeatability of the process and no significant differences were found even between the 50 nm and 100 nm masks. We have also checked the change of fill factor and etching rate as a function of grating height which is presented in Fig. 4.17. With the 10 minutes of etching time, the etching rates were 72.2 nm/min and 71.9 nm/min for the 200 nm and 500 nm grating periods, respectively. The etching rate was lower for the etching time of 2:30 minutes, because, the chamber strike pressure was 60 mTorr and it

TABLE 4.1

Four optimized recipes with the selectivity of above 35 approximately.

Recipe	Gas composition		RF (W)	ICP (W)	Pressure (mTorr)	He-bake (Torr)	DC bias (V)	Etch rate nm/min
	Gases	Flow (sccm)						
1	CHF ₃ /Ar	10/15	20	600	3	5	75	71.9
2	CHF ₃ /Ar	10/15	30	600	3	5	105	94.1
3	CHF ₃ /Ar	10/15	30	600	9	0	110	99.0
4	SF ₆ /Ar	10/15	20	600	9	0	40	83.8

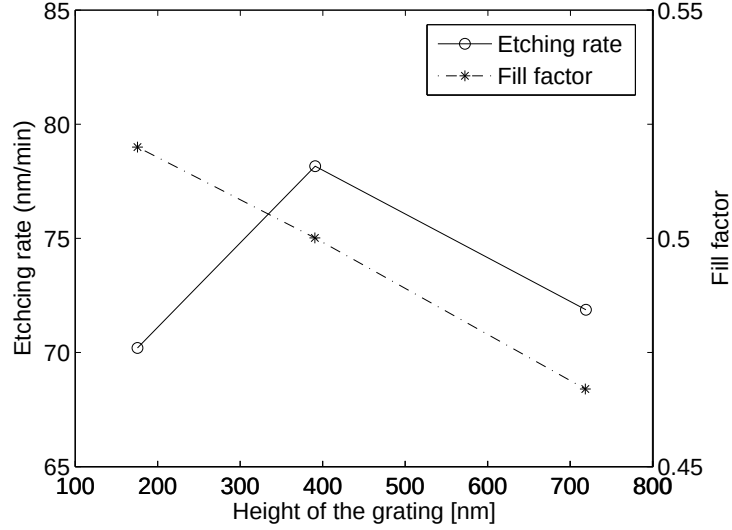


Figure 4.17: Etch rate and fill factor as a function of grating height (Recipe-1).

required around 30 seconds to reach the pressure of 3 mTorr. As we have discussed in Section 4.3 that the etching rate drops towards the higher chamber pressure. The fill factor decreased fairly with the higher grating height or the longer etching time. No significant loading effect was found for this optimized recipe. The SEM images of the both gratings etched with this recipe are shown in Fig. 4.13 (a, b) and 4.18 for the 10 minutes and 5 minutes of etching time, respectively. Where Fig. 4.13 (c) and (d) are the grating images, etched with the RIE for comparison. As we can see, thick layer of polymer was deposited on the sidewall of the gratings with the

RIE. This kind of polymer deposition was not desirable and with the aid of ICP our developed process resolves this issue remarkably.

Recipe-2 was developed to resolve the main limitation of the recipe-1 by increasing its RF power from 20 W to 30 W to etch the sample in the micrometer scale. We have discussed its characteristics in Section 4.7 and the example of the etched samples are already shown in Fig. 4.15. Recipe-3 is similar to the recipe-2 but it does not contain any He-backside cooling to reduce the polymer deposition on the sidewall. However, recipe-2 is still better in terms of flat and vertical sidewall with the least micro-trenching compared to this recipe. Therefore, we recommend to utilize the Recipe-3 only if the polymer deposition on the sidewall is problematic for any particular component. Fig. 4.19 represents the two SEM images of the both gratings, etched with the recipe-3. Recipe-4 is the best in terms of minimal micro-trenching as we have already seen in Fig. 4.16 (c). However, it has the limitation to work only for low aspect ratio features. For instance, the grating bars and grooves get triangular shape for the 200 nm period due to the tapered sidewall for 10 minutes of etching time.

In addition, we have checked some samples by cleaning with the O₂ plasma to remove the thin layer of polymer that deposit on the sidewall under the mask. The process parameters were: O₂ gas flow rate of 20 sccm, RF power of 100 W, chamber pressure of 60 mTorr, strike rate of 80 mTorr and process time of 5 minutes. Apparently the surface roughness of the sidewall reduced to some extent but no other significant differences were found. However, one can develop a process to remove this kind of deposited polymer from the sidewall.

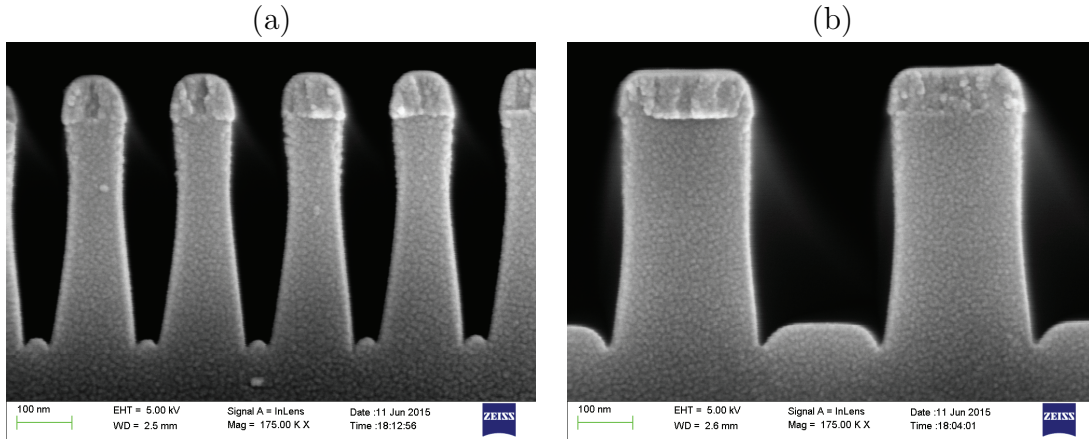


Figure 4.18: Cross-sectional SEM images of the gratings having periods of 200 nm (a) and 500 nm (b) etched with the gas flow rate of CHF_3/Ar 10/15 sccm, RF power of 20 W, ICP power of 600 W, chamber pressure of 3 mTorr, 5 Torr of He pressure (backside cooling) and etching time of 5 minutes.

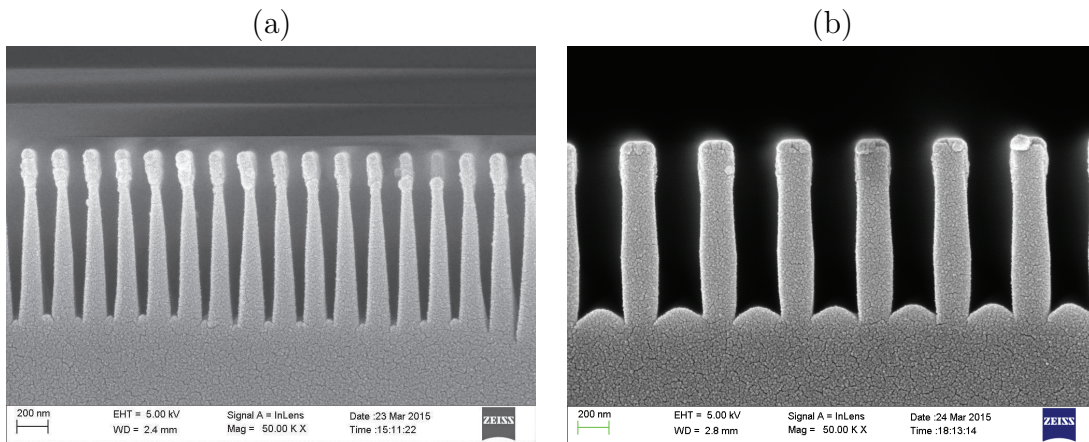


Figure 4.19: Cross-sectional SEM images of the gratings having periods of 200 nm (a) and 500 nm (b) etched with the gas flow rate of CHF_3/Ar 10/15 sccm, RF power of 30 W, ICP power of 600 W, chamber pressure of 9 mTorr, and etching time of 10 minutes. The etch rate of the process was 99 nm/min.

The etching characteristics of SiO_2 using ICP have been studied. The change of etch rate, selectivity and profile quality have been examined and compared against the change of RF power, ICP power, chamber pressure, gas flow rate and substrates backside cooling. The detailed fabrication method of the samples has been discussed. It was found that the different structure size limits the evaluation of the optimization of the process. Therefore, we have proposed four optimized processes which were aimed to etch different structure sized elements.

The Selectivity was found to be reasonable between RF power from 20 W to 30 W, when 600 W ICP power was applied. The enhanced anisotropic etch was observed with the chamber pressure of 3 mTorr. The chamber pressure more than 9 mTorr reduced the selectivity and anisotropy of the etch significantly. It was observed that more than 50% Ar reduces the etching rate gradually and the micro-trenching was minimal with the 60% Ar addition. Also the total gas flow rate of 25 mTorr was found to be feasible with the 60% Ar additive. It was observed that only 1 sccm of O_2 gas addition affects the profile quality significantly. It reduced the sidewall passivation and we have seen undercutting with the CHF_3 and the increased micro-trenching with the SF_6 reactive gases. However, the sidewall surface roughness was reduced with the O_2 additive. The He-backside cooling improved considerably the sidewall passivation and provided flat and vertical sidewall in the case of CHF_3 reactive gas. However, excessive polymer deposition occurred with the SF_6 reactive gas. The micro-trenching effect was the most challenging phenomenon to resolve and it was found out that it occurred due to the slanted sidewall of the mask. Therefore, to obtain a profile with minimal micro-trenching, we recommend

to improve the mask quality by developing its fabrication method.

The ultimate optimized process was developed to etch the submicron structures. The process parameters were: the CHF_3/Ar gas flow rate of 10/15 sccm, the RF power of 20 W, the ICP power of 600 W, the chamber pressure of 3 mTorr and the He pressure of 5 Torr (backside cooling). The etch rate of the process was 71.9 nm min^{-1} and the selectivity was well above 35. The process provides the best flat and vertical sidewall with the negligible micro-trenching. No significant loading effect was observed and the change of the fill factor against the grating height was also reasonable. Therefore, this process is also applicable to the multi-featured structures. However, a particular component might require fine-tuning of the optimized parameters. We have presented the effects of the associated parameters so that one can perform the fine-tuning of the optimized process further. In addition, the influence of chiller temperature, the effects of different mask materials and mask quality in ICP etching has to be investigated thoroughly. The optimized solutions also require further development for a deep shallow structure in the micrometer scale.

The optimization of an ICP-RIE system is an endless process. The stability and reproducibility of the etching characteristics are strongly dependent on the specific ICP-RIE system. On the other hand, the etch quality and accuracy varies depending onto the structure size of the components. Therefore, to obtain a precise profile structure, we highly recommend to perform fine-tuning of the provided optimized parameters for a particular SiO_2 component and ICP-RIE system.

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Used Tools and Equipment

- Kurt J. Lesker LAB18
- Headway spinner PWM101D
- Vistec EBG5000+ES HR
- OPTIspin SST20
- Plasmalab 100 plus
- Automatic dicing saw DAD3240
- Plasmalab 80 plus
- Emitech K675X
- Leo 1550 Gemini
- Ultrasonic cleaning machine
- Hot plate
- Glass cutter pen
- Nitrogen gun
- Vacuum oil
- Tweezers, scalpel, pipette, beakers and scotch tape.